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Multi-Modal Large Language Models are Effective Vision Learners

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Abstract

Large language models (LLMs), pre-trained on vast amounts of text, have shown remarkable abilities in understanding general knowledge and commonsense. Therefore, it's desirable to leverage pre-trained LLM to help solve computer vision tasks. Previous works on multi-modal LLM mainly focus on the generation capability. In this work, we propose LLM-augmented visual representation learning (LMVR). Our approach involves initially using a vision encoder to extract features, which are then projected into the word embedding space of the LLM. The LLM then generates responses based on the visual representation and a text prompt. Finally, we aggregate sequence-level features from the hidden layers of the LLM to obtain image-level representations. We conduct extensive experiments on multiple datasets, and have the following findings: (a) LMVR outperforms traditional vision encoder on various downstream tasks, and effectively learns the correspondence between words and image regions; (b) LMVR improves the generalizability compared to using a vision encoder alone, as evidenced by its superior resistance to domain shift; (c) LMVR improves the robustness of models to corrupted and perturbed visual data. Our findings demonstrate LLM-augmented visual representation learning is effective as it learns object-level concepts and commonsense knowledge.

1. Introduction

Visual representation learning aims to extract useful and meaningful representations from images. It is a fundamental task in computer vision and artificial intelligence, as it enables machines to understand and interpret the visual world. Visual representation learning can be done in various ways, including supervised and unsupervised approaches. The learned representation can capture the underlying structure, patterns, and features of the images, and make them easier to process and understand for various downstream tasks, including image classification [6], object detection [7], image segmentation [47] and others. There are two main approaches to visual representation learning, including supervised learning and unsupervised learning. Supervised methods typically use labeled datasets (e.g. ImageNet) to pre-train the model, and then use the features from the hidden layers for downstream tasks. For unsupervised or self-supervised methods, the models are pre-trained using unlabeled data. More specifically, these methods typically leverage heuristics-based pretext tasks (e.g., image inpainting, rotation prediction) to pre-train the model. While there are many previous work on visual representation learning, the learned features are not robust to common corruptions [35] (e.g. whether effect) and domain shift [20].

Figure 1. Example visualization of token-level attention map from multi-modal LLM. The leftmost panel shows the original image. We overlay the attention map between the word *broccoli* and *cup* in the generated caption and image patches in the right two panels. We found our model learns correspondence between word and image region. The learned multi-modal concepts can be helpful for solving downstream tasks. More details and full generated caption can be found in Section. 4.2.

Large language models (LLM) [32, 38] are neural network models trained on massive amounts of text data. By learning from text from different sources, LLM acquires general and commonsense knowledge that humans use to understand and interact with the world [36, 46]. In the natural language processing (NLP) domain, LLMs have achieved state-of-the-art performance in various downstream tasks [12, 25]. In order to utilize LLM to help solve computer vision tasks, there are recent works [23, 24] on multi-modal large language models, which aim to combine the power of pre-trained large language models (LLMs) and vision encoders. Multi-modal LLM typically uses LLM as

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the core component, and integrates with a vision encoder to perform various tasks involving both text and images. Multi-modal LLM can leverage the rich semantic and syntactic knowledge of LLMs, as well as the visual perception and understanding capability of image encoders. These models have achieved improved performance on a range of downstream tasks, such as image captioning, visual question answering, and story generation [22]. However, most of the existing work on multi-modal LLM focuses on generation tasks, and using LLM for visual representation learning is a relatively under-explored area.

In this work, we aim to leverage LLMs to improve the quality for visual representation learning. Specifically, we use LLM to provide additional prior knowledge about items in the image, and control feature extraction by supplying a text prompt as input. We hypothesize that LLM can enhance the extraction of visual features. First, LLM has the potential to learn more robust features, because LLM is trained on text corpora produced by human, it learns higher-level concepts. In addition, we believe LLM can help learn more generalizable visual features. Because LLM can leverage the rich semantic information from natural language to guide the visual feature extraction process. This can result in more meaningful and diverse features that can capture the essence of the visual data, rather than relying solely on low-level details.

To this end, we propose a novel method, LLM-augmented visual representation learning. Our approach seamlessly integrates a large language model (LLM) into visual representation learning, thereby enhancing the capacity to extract meaningful features from images. By aligning the output feature of the vision encoder with the word embedding space of LLM, we can leverage the prior knowledge in LLM to enhance visual feature extraction. Our code is publicly available at https://github.com/lisun-ai/LMVR.

Our contributions can be summarized as follows:

- We propose large language model augmented visual representation learning (LMVR), which leverages LLM to enhance feature extraction from images.
- We propose and benchmark various approaches for sequence-level aggregation of features extracted by LLMs.
- We conduct extensive experiments on multiple datasets, which demonstrate that LMVR outperforms the vision encoder on various downstream tasks, and effectively learns the correspondence between words and image regions.
- We find LMVR improves the generalizability compared to using a vision encoder alone, as demonstrated by its better resistance to domain shift.
- We find LMVR can also improve the robustness to corrupted and perturbed visual data.

2. Related work

2.1. Visual Representation Learning

Representation learning aims to map raw data (e.g. images) into a lower-dimensional latent space, where the learned representation can be useful for downstream tasks. There are two main approaches to visual representation learning, including supervised learning and unsupervised learning. For supervised learning methods, it's a common practice to pre-train the models on a large-scale annotated dataset (e.g. ImageNet [11]), and the learned features can be transferred to target tasks that often have less training data. However, there are two limitations. First, obtaining a large amount of annotated data is not always available since it is expensive and time-consuming to collect labels. In addition, the learned representation may not generalize well to new or unseen data, especially if the data distribution changes or the downstream task is different from the training task. Unsupervised learning methods can be categorized into discriminative and generative approaches. For discriminative methods, early approaches utilize heuristics-based pretext tasks (e.g. solving jigsaw puzzles [26] and rotation prediction [16]). Contrastive learning has been recently proposed as a more principled variant of self-supervised learning. It relies on instance discrimination [42] or contrastive predictive coding (CPC) [27], and has achieved great success on many downstream tasks [34,37,43–45]. The primary objective of contrastive learning is to bring together similar data instances while pushing apart dissimilar ones in a shared feature space [6]. By doing so, it learns to extract embeddings that capture semantic information. For multi-modal representation learning, CLIP [31] introduces contrastive learning between images and associated captions. Therefore, it learns to project images and their textual descriptions in a shared feature space.

Generative approaches learn representations by modeling the underlying data distribution. Early methods adopt heuristics-based pre-training tasks, including image inpainting [29] and denoising auto-encoder [40]. Variational auto-encoder [21] is another popular latent variable model for representation learning. VAE consists of an encoder network that maps the input data to a latent space, and a decoder network that reconstructs the input data from the latent space. The latent space is usually assumed to follow a simple prior distribution, such as a Gaussian distribution. More recently, masked auto-encoders (MAE) [17] and BEiT [4] are proposed. Both of them adopt masked image modeling as pre-training tasks. MAE conducts the reconstruction on the pixel level, and BEiT performs the reconstruction on the token level. Despite the progress, most

current methods only use a vision encoder to extract features, without leveraging external sources of prior knowledge, such as a pre-trained LLM.

2.2. LLMs and Multi-Modal LLMs

Large Language Models (LLMs) stand out for their capacity to attain comprehensive language comprehension and generation, showcasing their capability to handle general-purpose language tasks. Traditional language models are often limited by their vocabulary size, context length, and model capacity. Thus, they may not be able to capture the rich and diverse information contained in natural language, such as world knowledge. LLMs are designed to overcome these limitations by using efficient and scalable architecture (mainly Transformers [39]) that are pre-trained using self-supervised objectives on unprecedented amounts of data. Pre-trained LLMs achieve state-of-the-art performance on many downstream tasks [2, 28].

In order to handle tasks that require multi-modal reasoning and understanding, multi-modal LLMs integrate LLMs with visual encoders that can process and represent visual data. Multi-modal LLMs can leverage the powerful language capabilities of LLMs and the visual perception capabilities of visual encoders to perform multi-modal tasks. For example, LLaVA [24] connects pre-trained CLIP [31] vision encoder with the LLaMA model [38]. BLIP-2 [22] introduces a query Transformer to bridge frozen EVA vision encoder [14] with a LLM. Flamingo [1] introduces a Perceiver Resampler that produces a fixed number of visual tokens using features extracted from a vision encoder. Next, the visual tokens are used to condition the frozen LLM using a cross-attention layer. LLaMA-Adapter V2 [15] first projects features extracted by the vision encoder into text domain, then injects them into the early LLM layers of LLaMA. It only fine-tune bias parameters in LLM. Multi-modal LLMs are typically trained using multi-modal data sources, such as text-image pairs, text-video pairs, or text-audio pairs, which can enrich their knowledge and improve their generalization. Multi-modal LLM has achieved great success in many multi-modal downstream tasks, including image captioning, visual question-answering, and others [8, 41]. However, these efforts mostly focus on generative tasks. Leveraging LLMs for visual representation learning is a relatively under-explored area.

3. Method

In this section, we introduce our proposed method. Traditional methods only utilize a vision encoder to learn semantic visual embedding through supervised or unsupervised approaches. While these methods are empirically successful, the learned representation is not robust to image corruption and domain shift [3, 35]. For example, while humans can recognize images even when corrupted or degraded, vision encoder-extracted features are vulnerable to common perturbation (e.g., weather effect) [18]. In addition, the extracted image embedding may contain redundant information irrelevant to a specific downstream task.

3.1. LLM-Augmented Visual Representation Learning

In this work, we propose LLM-augmented visual representation learning (LMVR), which leverages a pre-trained language model for visual representation learning. Specifically, our method includes four components. First, we use a vision encoder (e.g., CLIP [31] vision tower) to extract representation from the image. Second, in order to connect the vision encoder with LLM, we use a projector (e.g., MLP) to transform features from vision encoder output space to LLM word embedding space. Next, we employ an LLM to take projected vision features and a text prompt as input, and generate an output. The prompt conditioning allows us to query the information in the embedding that's useful for different downstream tasks. Our method is agnostic to the choice of vision encoder and LLM. Finally, we design a sequence-level aggregation method that integrates token-level representation into an image-level representation, which is used in downstream tasks. An illustration of our method is shown in Fig. 2.

3.2. Sequence-Level Aggregation

As discussed in Section. 3.1, the large language model produces a sequence of tokens. In order to have a compact image-level representation, we design and benchmark different sequence-level aggregation methods. Specifically, we studied the following different design choices for extracting representation from LLM:

- Extract embedding from single token v.s. Aggregate many tokens: While most previous work use embedding from single `[CLS]` or `[END]` token, we also studied aggregating embedding from all generated tokens or the entire token sequence, including input tokens and output (generated) tokens.

- Aggregation method: Average pooling v.s. Max pooling: For the token embedding aggregation method, we implement and compare both average pooling and max pooling. We choose average pooling because it makes sense to obtain sentence-level embedding by averaging representation from all tokens. We also benchmark max pooling because it captures the most salient feature in the sentence.

- Features from last hidden layer v.s. Previous layers: While most previous methods use feature representation from the last hidden layer, we also benchmark the performance of hidden states from earlier layers.

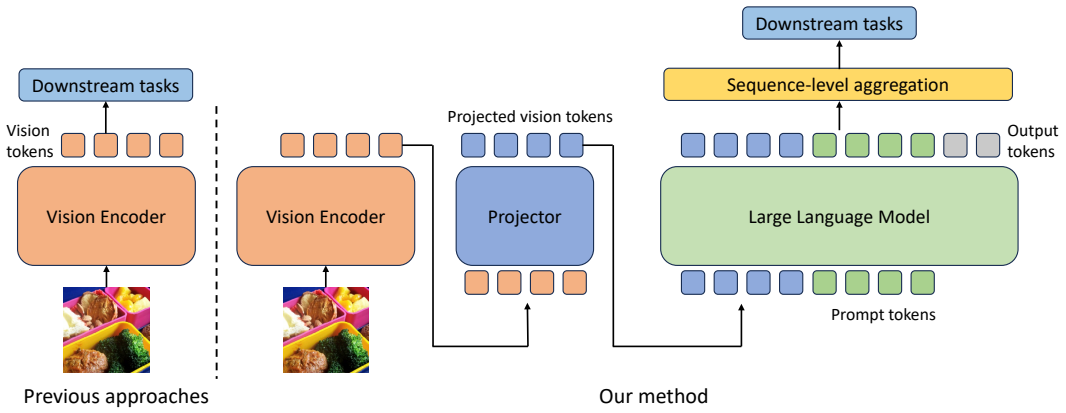

Figure 2. Illustration of our approach. Previous methods only use a vision encoder for representation learning. Our method is comprised of four steps. First, a vision encoder extracts features from images. Then, the extracted features are projected into the space of word embedding. Next, the LLM takes the projected visual features and a text prompt as input and generates a response. Finally, we perform sequence-level aggregation of features of hidden layers to obtain image-level representation.

- Extract features before generation v.s. after generation: We compare the quality of features extracted before letting LLM decode to generate output, which only uses the encoding ability of LLM, and features extracted after LLM generation, which requires more computing but will leverage the generation power of LLM.

We found that the average pooling of all output tokens achieves the best results. We present the detailed benchmark settings in Section. 4.1.1.

3.3. Training

We use the vision tower of CLIP [31] (ViT-large) as the vision encoder, and Vicuna-7B [9] as LLM, which is fine-tuned on LLaMA [38]. We choose these components because they have demonstrated outstanding performance in visual and language understanding tasks, respectively, and have been widely adopted in previous vision-language models. We follow the practice of LLaVA [24] to pre-train our model in two stages. During the first stage, we keep both the visual encoder and LLM weights frozen, and only pre-train the projector. We use 595K image-text pairs from filtered CC3M dataset [5]. The image and a generic prompt (e.g. *Provide a brief description of the given image*) are used as input. Then, the LLM is prompted to generate a response. The negative log-likelihood of predicting the target caption is used as the loss function. During the second stage, we fine-tune both the projector and the LLM. We use the instruction tuning data from LLaVA [24]. The negative log-likelihood of predicting the target response is used as the loss function. We train our model for one epoch during the first stage for feature alignment, and three epochs for the second stage for visual instruction tuning. The first stage of feature alignment ensures that the visual and textual modalities are mapped to a common latent space, which is crucial for multi-modal tasks. This alignment improves the coherence between modalities, enabling the model to better understand visual content in relation to text. The second stage of visual instruction tuning further refines this process by guiding the model to focus on specific aspects of the visual data based on the provided instructions. Together, these components enhance the quality of the extracted features by making them more semantically meaningful and task-relevant.

4. Experiments

We perform an extensive evaluation of extracted representation in terms of semantic quality, robustness, and generalizability.

4.1. Benchmark Visual Representation

In this section, we first benchmark different design choices for sentence-level embedding aggregation of our method. Then, we conduct a component-wise comparison to quantitatively study the performance gain brought about by introducing LLM. Finally, we present an object-level visualization.

4.1.1 Explore the Best Practice for Feature Extraction from LLM

In order to find the optimal approach for sequence-level aggregation, we conduct experiments on the classification task of the VOC07 dataset [13]. The VOC07 dataset contains multi-class labels for 20 object classes. Following the practice of CLIP [31], we freeze our pre-trained model, and train a linear classifier on top of extracted features, then evaluate the performance on classification accuracy. The results

are shown in Table. 1. We observe the following insights: First, we found that aggregating embedding of all generated tokens achieves better performance than only using the embedding of the [END] token. We think this is because aggregating multiple tokens will maintain more information than only using a single token. Second, in terms of the aggregation method, we found average pooling is better than max pooling. We hypothesize that average pooling preserves more information about the input image. In addition, we found that embedding from the last layer slightly outperforms previous layers. The best approach we found is to perform average pooling of all output tokens.

Table 1. Benchmark the design choices in feature extraction with LLM on VOC07 dataset.

Method	mAP (%)
Only use embedding from last [END] token	90.5
Max pooling of embedding of all generated tokens	92.2
Average embedding of all generated tokens	93.5
Average embedding from all output tokens (second-to-last layer)	93.7
Average embedding from all output tokens	**93.8**

4.1.2 Component-Wise Comparison

Large language models that are learned from massive amounts of text data have the ability to capture general knowledge and common sense, which can be beneficial for extracting visual features from images. To quantitatively study the performance gain by introducing LLM, we conducted a component-wise comparison. In detail, we extract visual features from the vision encoder, the projector, and the LLM, respectively. Then, we perform linear probe evaluation on different datasets. Shown in Table. 2, we found that LLM-extracted visual features demonstrated a 2.5% improvement in mean Average Precision (mAP) on the VOC07 multi-label classification task compared to the vision encoder alone. LLM-extracted visual features also lead to 1% mAP improvement on the COCO dataset. These results highlight the potential for leveraging pre-trained LLMs to improve the performance of multi-modal models in various visual tasks. Hence, our LMVR performs better on tasks of multi-classification, such as VOC07 and COCO. We believe this is because LLM acquires general knowledge beforehand, which enables it to comprehend complex scenes.

4.2. LMVR Learns Object-Level Concepts

We hypothesize that the performance gain of incorporating LLM for feature extraction is due to our method learns object-level multi-modal concepts. To validate it, we visualize the text-to-image attention map from the LLM decoder. Specifically, we feed our model a test image and a prompt (*Describe the items in the image*), then we use the model to generate a response. For the example in Fig. 1 and Fig. 4, the generated responses are available in the Supplementary Material. Next, we visualize the attention map of how the noun tokens in the generated response attend to different patches in the input image. We use the attention map of the middle layer (17th Transformer layer out of 32) of LLaMA-7B. Shown in Fig. 4, we found that the attention map correctly highlights the region that corresponds to the query words, revealing that LLM learns object-level multi-modal concepts during unsupervised pre-training, which is helpful for solving downstream tasks.

Table 2. Evaluation of component-wise comparison. We use the mean average precision (mAP) as the evaluation metric.

	VOC	COCO
Vision encoder (CLIP)	91.3%	79.5%
w/Adaptor	91.0%	78.5%
w/LLM	**93.8%**	**84.0%**

4.3. LMVR Learns More Generalizable Visual Representation

It is desirable to have learned visual representations that are generalizable, which means that they can perform well on new or unseen data without requiring additional training or fine-tuning. LLM is trained on huge amount of text corpus, the learned knowledge can be transferred to different downstream tasks, thus LLM has the potential to help improve the generalizability of learned visual representation. In order to study how well the visual representation extracted using our LMVR can transfer from one task or domain to another, we conduct experiments on the VisDA 2017 dataset [30]. In this dataset, the training set contains synthetic images from 3D CAD models, while the test set contains real images from YouTube BBox datasets [33]. The classification task has 12 object categories, including airplane, horse, and others.

In order to study whether LLMs can help improve the generalization of extracted visual representation, we train a linear classifier on the features extracted from the training set of CAD images, then test this classifier on the test set of natural images. Shown in Table. 3, we found that features extracted from LLMs achieve a large 18.2% performance gain compared with using the image encoder alone under domain shift. We believe this performance gain is due to that multi-modal LLM focuses more on object-level features than the image encoder alone.

In order to understand why LLMs can help improve the generalization of visual features, we take some sample images from different domains, and feed them with a prompt

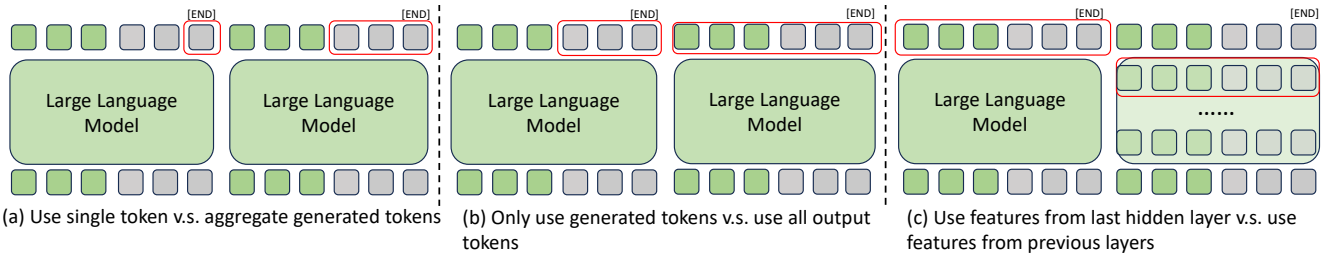

Figure 3. Schematic of design choice tested to study the best strategy for extracting hidden layer features from LLM. The gray rectangle indicates tokens in the generated response. The red rectangle indicates the set of tokens used for feature extraction.

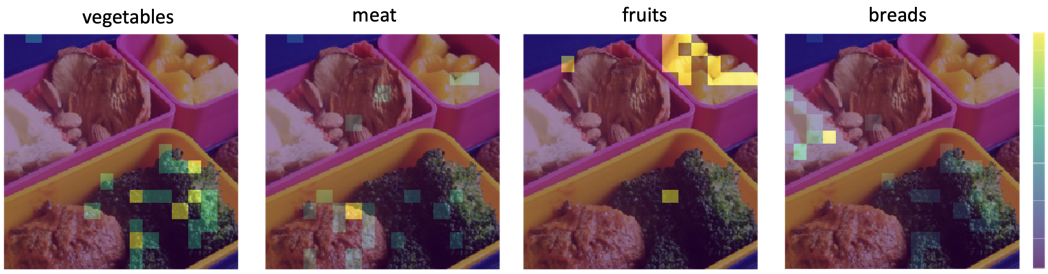

Figure 4. Results of token-level visualization. We overlay a sample image with attention maps between selected text tokens in generated response and vision tokens. The full generated response can be found in Section. 4.2. We discover that the attention map accurately highlights the area that matches each query word, showing that our model grasps multi-modal concepts at the object level.

Table 3. Evaluation results on VisDA test set

Method	Accuracy
Vision encoder (CLIP)	60.9%
LMVR (Ours)	**79.1%**

Describe the items in the image into LLM to generate a description. As shown in Fig. 5, we found that despite the images coming from different domains and having different visual characteristics, the LLM-generated response still accurately captures the category.

4.4. LMVR Learns More Robust Visual Representation

In real-world downstream applications, models may face various forms of input image degradation, such as noise, blur, occlusion, and distortion. Therefore, robustness to corruptions and perturbations is desirable for visual representation learning. In order to study the robustness of the learned representation from LMVR, we conduct experiments on the ImageNet-C dataset [18], which introduces different types of corruptions to the ImageNet validation set, including weather effects, digital artifacts, camera defects, and physical transformations.

We extract embeddings from the vision encoder and LLM from our model, respectively, and conduct a linear probe experiment to compare the quality of these two visual representations under different types of corruptions. As shown in Table. 4, we found that with the help of the LLM, the extracted visual features are more robust to corruption and perturbation (4.7% average improvement in accuracy) than using just the image encoder alone. In addition, we report the average performance for each perturbation category. The results are shown in Table. 5. Our LMVR outperforms the vision encoder in every category. The performance gap is the largest for blurring. We believe the reason is that blurring mostly removes high-frequency details, and the vision encoder is sensitive to it. Since LLM focuses more on high-level object-related features, the extracted representation is more robust to blurring.

In order to understand why LLMs can help improve the robustness of visual feature extraction, we use LLMs to generate descriptions for clean test images and their corrupted versions. As shown in Fig. 6, we found that even under moderate corruption, LLM-generated responses can still accurately capture the main objects in the image.

4.5. LMVR Boosts the Comprehension of Objects, Instead of Low-Level Textures

A typical image contains both high-level objects and low-level textures. High-level objects are more conceptual and thematically significant features of an image, such as items, scenes, and interactions. Low-level textures are more basic and superficial features of an image, such as colors, edges, and shapes. In previous sections, we show that our

LLM description:

The image features a large grey airplane with a pointy nose, sitting on top of a white paper plate. The airplane occupies a major portion of the scene, while the paper plate appears to be partially transparent and covers almost the entire base of the image. The plane seems to be a model, possibly a toy, due to its size and prominence in the image. There are no other objects or people visible in this scene.

LLM description:

The image shows a red fighter jet flying in a blue sky, with a few wispy clouds scattered across the sky. The jet appears to be at full throttle, and it is heading straight up into the sky.

Figure 5. Example of LLM generated responses on VisDA dataset. The results show that the LLM-generated description can identify the key objects in the images, as marked in red, even when the images belong to different domains.

Clean

LLM description:

In the image, there is a young man kneeling down while holding a large fish in his hands. The fish is lying on its side, and the boy appears to be proud of his catch. The surroundings include a grassy area with green grass and trees in the background, suggesting that the boy might have been fishing in a nearby body of water.

Gaussian Blur

LLM description:

In the image, there is a man standing in a grassy area holding a large fish in his hands. The fish appears to be a bass, and it is quite a sizeable catch. The man is posing with the fish, likely displaying his skills as a fisherman or simply enjoying the moment.

Gaussian Noise

LLM description:

In the image, there is a young man kneeling down next to a large fish. The fish appears to be bass-like with its mouth wide open. The young man is posing with the fish, showcasing its size and catching a memorable moment. The scene likely takes place near a body of water where the fish was caught.

Figure 6. Example of LLM generated responses under image corruption. The images are from the ImageNet-C dataset. We found even under a moderate amount of noise, the generated response from LLM is robust and can still capture the gist of the image, as highlighted in red. To preserve privacy, the person's face is masked.

method can help learn more generalizable and robust visual features. We hypothesize the reason is that compared to vision encoders, LLM focuses more on high-level features rather than low-level features. In order to validate it, we conduct experiments on the DeepFashion-Multimodal dataset [19]. The dataset contains high-resolution human images, as well as the annotation for outfit categories, fabric, and texture of clothes. Following the setting in Section. 4.1.2, we first use our pre-trained model to extract features from the images. We use a generic text prompt (*Describe the items in the image*) as input. Next, we train a linear classifier on top to predict the label categories. The results are shown in Table. 6. We found that LLM provides performance gain mainly on the object classification task. For tasks that only require low-level understanding, including texture and texture classification, LLM doesn't bring a significant advantage. We hypothesize that this is because during pre-training, LLM mainly learns abstract high-level concepts; thus, it may not provide performance gains for low-level textures.

Our model is trained on a generic dataset of natural images. To examine the effect of fine-tuning for a domain-specific task, we fine-tune the LLM in our model using the images and associated text description in the DeepFashion-Multimodal dataset. The results are shown in Table. 6. We found that fine-tuning with domain-specific data can further improve the performance, and the gain is mostly from object classification, not low-level details such as fabric and texture.

4.6. Generalizability to Other Architectures

In the sections above, we conduct experiments with LLaVA-style architecture, which uses a linear layer as the projector and the Vicuna model as the LLM component. In order to study the generalizability of our approach to different architectures, we adapt our model with BLIP-2 [22]. In detail, the model uses EVA [14] as the vision encoder, Q-former as the projector, and uses FlanT5-XXL [10] as the

Table 4. Evaluation results on ImageNet-C dataset. Our proposed LMVR is more robust to different types of corruption compared to using an image encoder alone.

Accuracy(%)	Weather				Noise			Blur				Digital				
Methods	Brightness	Fog	Frost	Snow	Gaussian	Impulse	Shot	Defocus	Glass	Motion	Zoom	Contrast	Elastic	JPEG	Pixelate	Average
Vision encoder (CLIP)	79	67	56	65	62	57	60	49	36	60	49	67	69	69	68	61
LMVR (Ours)	**81**	**71**	**57**	**67**	**64**	**61**	**62**	**62**	**45**	**68**	**54**	**75**	**71**	**74**	**75**	**66**

Table 5. Evaluation results on ImageNet-C dataset, averaged by perturbation category

Accuracy(%)	Weather	Noise	Blur	Digital
Vision encoder (CLIP)	66.9	59.7	48.6	68.0
LMVR (Ours)	**69.0**	**62.3**	**57.1**	**73.7**

Table 6. Evaluation results on DeepFashion-Multimodal dataset

Accuracy (%)	Object	Fabric	Texture
Vision encoder (CLIP)	89.1	73.8	71.3
LMVR	90.0	**73.9**	**71.4**
LMVR (fine-tuned)	**91.2**	**73.9**	71.2

LLM component. First, we conduct experiments to compare the performance between our method and only using the EVA vision encoder. The results on VOC07 and COCO datasets are shown in Table. 7. We found that our method outperforms the vision encoder by a decent margin.

Table 7. Evaluation of our method with BLIP-2 on VOC07 and COCO datasets

mAP	VOC	COCO
Vision encoder (EVA)	92.2%	82.3%
LMVR (Ours)	**93.7%**	**84.5%**

In addition, we study the robustness of extracted visual features against domain shift. Following the experiment settings in Section. 4.3, we conduct linear probe experiments on VisDA [30] dataset, and compare the performance between only using the vision encoder and our LMVR. We use the same text prompt (*Describe the items in the image*). The training set contains synthetic images from 3D CAD models. The test set consists of real images from YouTube BBox datasets. The results are shown in Table.8. We found that LMVR outperforms the vision encoder by 8.9%, which demonstrates that with the help of LLM, the learned representation has better transferability. The results in this section indicate that our method is generalizable to different vision encoders and LLM architectures.

Table 8. Evaluation results on VisDA test set with BLIP-2

Method	Accuracy
Vision encoder (EVA)	72.1%
LMVR (Ours)	**81.0%**

5. Discussion

We think there are two main advantages of leveraging multi-modal LLM for visual feature extraction. First, multi-modal LLM focuses more on abstract object-level features. While the conventional approach of using the vision encoder alone may rely on low-level textures. We think that the reason is that LLM is pre-trained on a large amount of human-generated text corpora. In this way, the learned representation will align more with human's understanding and perception of diverse objects and concepts. This property is helpful as it encourages the model to learn more robust and adaptable features. Second, LLM learns commonsense knowledge during pre-training. The prior knowledge can be beneficial for downstream tasks that require the understanding of complex scenes. In addition, we would like to note our token-level attention map provides a visual understanding of how the model distributes focus across different parts of the image. Although in most cases the attention map highlights the area that matches each query word, the highlighted area may not cover the entire object. Since the pre-training objective focuses on predicting captions from images, rather than performing supervised object segmentation, the model is not explicitly trained to identify precise object boundaries. As such, it is important to view these attention maps as interpretive tools for the model's inner workings.

6. Conclusion

In this work, we propose large language model augmented visual representation learning (LMVR), which leverages pre-trained LLM for improved visual representation learning. Future directions of our work include designing a better alignment method between image and text modalities, and extending to other modalities (e.g. audio and video). In addition, prompt engineering can be explored for better and task-specific visual feature extraction.

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